

ERASMUS UNIVERSITY ROTTERDAM

ERASMUS SCHOOL OF ECONOMICS

Master's Thesis Econometrics and Management Science - Business Analytics and Quantitative
Marketing

Median-of-Means Two-Stage Least Squares Regression

David von Schledorn Márquez (597334)



Supervisor:	Mikhail Zhelonkin
Second assessor:	-
Date final version:	-

The views stated in this thesis are those of the author and not necessarily those of the supervisor, second assessor, Erasmus School of Economics or Erasmus University Rotterdam.

Abstract

Median-of-Means methods are used for instrumental variable regression and their properties are compared to the standard 2-stage least squares method.

1 Preliminaries

1.1 Setup and Assumptions

Let $(Y_i, X_i, Z_i)_{i=1}^n$ be an i.i.d. random sample from a joint distribution on $\mathbb{R}^3$ satisfying the structural equation

$$Y_i = \beta X_i + \varepsilon_i, \quad (1)$$

where $\beta \in \mathbb{R}$ is the structural parameter of interest and ε_i is an unobserved error term. We impose the following conditions.

(A1) **Exogeneity.** $\mathbb{E}[Z_i \varepsilon_i] = 0$.

(A2) **Relevance.** $\mu_{ZX} := \mathbb{E}[Z_i X_i] \neq 0$.

(A3) **Finite second moments.** $\sigma_{ZY}^2 := \text{Var}(Z_i Y_i) < \infty$, $\sigma_{ZX}^2 := \text{Var}(Z_i X_i) < \infty$, and $\sigma_{Z\varepsilon}^2 := \text{Var}(Z_i \varepsilon_i) < \infty$.

Define $\mu_{ZY} := \mathbb{E}[Z_i Y_i]$. By (A1) and (A2), the structural parameter is identified as $\beta = \mu_{ZY} / \mu_{ZX}$.

Remark 1.1 (Moment conditions). *The conditions (A1)–(A3) hold for any distribution with finite second moments of the products $Z_i Y_i$, $Z_i X_i$, and $Z_i \varepsilon_i$. Thus, no assumptions are imposed on the marginal distributions of Y_i , X_i , Z_i , or ε_i .*

Remark 1.2 (Relationship between variances). *The three variances in (A3) are related. Since $Z_i Y_i = \beta Z_i X_i + Z_i \varepsilon_i$, Minkowski's inequality gives*

$$\sigma_{Z\varepsilon} \leq \sigma_{ZY} + |\beta| \sigma_{ZX}. \quad (2)$$

To see this, set $U = Z_i Y_i$ and $V = \beta Z_i X_i$, so that $U - V = Z_i \varepsilon_i$. By Minkowski's inequality for $p = 2$,

$$\sqrt{\text{Var}(U - V)} \leq \sqrt{\text{Var}(U)} + \sqrt{\text{Var}(V)},$$

which gives $\sigma_{Z\varepsilon} \leq \sigma_{ZY} + |\beta| \sigma_{ZX}$.

1.2 Sub-Gaussian Estimators

Definition 1.3 (Sub-Gaussian Estimators). *Let $(X_i)_{i=1}^n$ be a random sample with $\mathbb{E}[X_i] = \mu$ and $\text{Var}[X_i] = \sigma^2$, and let $\bar{\mu}(X)$ denote an estimator for μ . For some constant $L > 0$, $\bar{\mu}(X)$ is said to be a sub-Gaussian estimator for μ if*

$$\mathbb{P} \left[|\bar{\mu}(X) - \mu| > L\sigma \sqrt{\frac{\ln(1/\delta)}{n}} \right] \leq \delta \quad \text{for all } \delta \in (0, 1). \quad (3)$$

Make
clearer

1.3 The Median-of-Means Estimator for a Scalar Mean

We present the scalar Median-of-Means (MoM) estimator and its concentration properties. The results in this section are adapted from Devroye, Lerasle, Lugosi and Oliveira (2016) and Lugosi and Mendelson (2019).

Algorithm 1: Scalar Median-of-Means Estimator

Input: Sample $(X_i)_{i=1}^n$, number of blocks k
Output: Estimate $\hat{\mu}$
 Partition $(X_i)_{i=1}^n$ into k blocks $B_1, \dots, B_k$ of size $m = \lfloor n/k \rfloor$;
for $j = 1, \dots, k$ **do**
 Compute the block mean $\bar{\mu}_j = \frac{1}{m} \sum_{i \in B_j} X_i$;
return $\hat{\mu} = \text{med}(\bar{\mu}_1, \dots, \bar{\mu}_k)$;

Theorem 1.4 (MoM concentration). *Let $(X_i)_{i=1}^n$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$. Let k be a positive integer, $m = \lfloor n/k \rfloor$, and $\hat{\mu}_{\text{MoM}}$ the MoM estimator with k blocks. Then*

$$\mathbb{P} \left[|\hat{\mu}_{\text{MoM}} - \mu| > \frac{2\sigma}{\sqrt{m}} \right] \leq e^{-k/8}. \quad (4)$$

Setting $k = \lceil 8 \ln(1/\delta) \rceil$ for $\delta \in (0, 1)$, this becomes

$$\mathbb{P} \left[|\hat{\mu}_{\text{MoM}} - \mu| > \sigma \sqrt{\frac{32 \ln(1/\delta)}{n}} \right] \leq \delta, \quad (5)$$

which is sub-Gaussian in the sense that the dependence on δ is logarithmic.

Proof. The proof proceeds in three steps.

Step 1 (Chebyshev on each block). Each block mean satisfies $\mathbb{E}[\bar{X}_j] = \mu$ and $\text{Var}(\bar{X}_j) = \sigma^2/m$. By Chebyshev's inequality with threshold $2\sigma/\sqrt{m}$,

$$\mathbb{P} \left[|\bar{X}_j - \mu| > \frac{2\sigma}{\sqrt{m}} \right] \leq \frac{\sigma^2/m}{4\sigma^2/m} = \frac{1}{4}.$$

Step 2 (Counting argument for the median). Define $\zeta_j = \mathbf{1}\{|\bar{X}_j - \mu| > 2\sigma/\sqrt{m}\}$. By Step 1, $\mathbb{E}[\zeta_j] \leq 1/4$. Since the blocks are disjoint subsets of an i.i.d. sample, the $(\zeta_j)_{j=1}^k$ are independent.

If $|\hat{\mu}_{\text{MoM}} - \mu| > 2\sigma/\sqrt{m}$, the median exceeds $\mu + 2\sigma/\sqrt{m}$ or falls below $\mu - 2\sigma/\sqrt{m}$. In either case, at least $k/2$ of the block means satisfy $|\bar{X}_j - \mu| > 2\sigma/\sqrt{m}$, since the median of k numbers can only lie outside an interval if fewer than half the numbers lie inside it. Therefore,

$$\mathbb{P} \left[|\hat{\mu}_{\text{MoM}} - \mu| > \frac{2\sigma}{\sqrt{m}} \right] \leq \mathbb{P} \left[\sum_{j=1}^k \zeta_j \geq \frac{k}{2} \right].$$

Step 3 (Hoeffding's inequality). The sum $\sum_{j=1}^k \zeta_j$ consists of k independent random variables with $\zeta_j \in [0, 1]$ and $\mathbb{E}[\sum_j \zeta_j] \leq k/4$. For $\sum_j \zeta_j$ to reach $k/2$, it must exceed its mean by

at least $k/2 - k/4 = k/4$. By Hoeffding's inequality (Hoeffding, 1963),

$$\mathbb{P} \left[\sum_{j=1}^k \zeta_j - \mathbb{E} \left[\sum_j \zeta_j \right] \geq \frac{k}{4} \right] \leq \exp \left(\frac{-2(k/4)^2}{k} \right) = e^{-k/8}.$$

This establishes (4). The sub-Gaussian form (5) follows by setting $k = \lceil 8 \ln(1/\delta) \rceil$ and noting $m \geq n/(2k)$ for $n \geq 2k$. $\square$

2 The Standard Instrumental Variable Estimator

Definition 2.1 (Standard IV estimator). *Consider the structural equation (1). The standard IV estimator $\hat{\beta}_{\text{IV}}$ is defined by*

$$\hat{\beta}_{\text{IV}} = \frac{\bar{S}_{ZY}}{\bar{S}_{ZX}}, \quad \bar{S}_{ZY} = \frac{1}{n} \sum_{i=1}^n Z_i Y_i, \quad \bar{S}_{ZX} = \frac{1}{n} \sum_{i=1}^n Z_i X_i.$$

Lemma 2.2 (Error decomposition). *Under (1) and (A1),*

$$\hat{\beta}_{\text{IV}} - \beta = \frac{\bar{S}_{Z\varepsilon}}{\bar{S}_{ZX}}, \tag{6}$$

where $\bar{S}_{Z\varepsilon} = \frac{1}{n} \sum_{i=1}^n Z_i \varepsilon_i$ satisfies $\mathbb{E}[\bar{S}_{Z\varepsilon}] = 0$.

Proof. Multiply both sides of $Y_i = \beta X_i + \varepsilon_i$ by Z_i and average:

$$\bar{S}_{ZY} = \frac{1}{n} \sum_{i=1}^n Z_i (\beta X_i + \varepsilon_i) = \beta \bar{S}_{ZX} + \bar{S}_{Z\varepsilon}.$$

Rearranging and dividing by $\bar{S}_{ZX}$ gives $\hat{\beta}_{\text{IV}} = \beta + \bar{S}_{Z\varepsilon}/\bar{S}_{ZX}$, which is (6). By (A1), $\mathbb{E}[Z_i \varepsilon_i] = 0$, so $\mathbb{E}[\bar{S}_{Z\varepsilon}] = 0$ by linearity of expectation. $\square$

Theorem 2.3 (Standard IV concentration). *Assume (A1)–(A3). For any $\delta \in (0, 1)$, if*

$$n \geq \frac{8\sigma_{ZX}^2}{\delta\mu_{ZX}^2}, \tag{7}$$

then

$$\mathbb{P} \left[\left| \hat{\beta}_{\text{IV}} - \beta \right| > \frac{2\sqrt{2}\sigma_{Z\varepsilon}}{|\mu_{ZX}| \sqrt{\delta n}} \right] \leq \delta. \tag{8}$$

Proof. By Lemma 2.2, the estimation error equals the ratio $\bar{S}_{Z\varepsilon}/\bar{S}_{ZX}$. To bound this ratio we need two things simultaneously: the denominator must be bounded away from zero, and the numerator must be small. We control each requirement with a separate probability event and then combine them via the union bound.

Step 1 (Two-event decomposition). For a denominator threshold $D \in (0, |\mu_{ZX}|)$ and an error threshold $t > 0$, both to be chosen below, define

$$A(D) = \{|\bar{S}_{ZX}| \geq D\}, \quad B(D, t) = \{|\bar{S}_{Z\varepsilon}| \leq tD\}.$$

On $A(D) \cap B(D, t)$, the denominator satisfies $|\bar{S}_{ZX}| \geq D$ by definition of $A(D)$, so

$$\left| \hat{\beta}_{IV} - \beta \right| = \frac{|\bar{S}_{Z\varepsilon}|}{|\bar{S}_{ZX}|} \leq \frac{tD}{D} = t.$$

Hence $\mathbb{P}\left[\left|\hat{\beta}_{IV} - \beta\right| > t\right] \leq \mathbb{P}[A(D)^c] + \mathbb{P}[B(D, t)^c]$ by the union bound. For the remainder of this proof we set $D = |\mu_{ZX}|/2$, the symmetric choice that yields the cleanest constants.

Step 2 (Chebyshev bounds). We bound each failure probability in turn.

The failure event $A^c = \{|\bar{S}_{ZX}| < |\mu_{ZX}|/2\}$ implies a large deviation of $\bar{S}_{ZX}$ from its mean. Specifically, if $|\bar{S}_{ZX}| < |\mu_{ZX}|/2$ then, by the reverse triangle inequality,

$$|\bar{S}_{ZX} - \mu_{ZX}| \geq |\mu_{ZX}| - |\bar{S}_{ZX}| > |\mu_{ZX}| - \frac{|\mu_{ZX}|}{2} = \frac{|\mu_{ZX}|}{2}.$$

Hence $A^c \subseteq \{|\bar{S}_{ZX} - \mu_{ZX}| > |\mu_{ZX}|/2\}$. Since $\text{Var}(\bar{S}_{ZX}) = \sigma_{ZX}^2/n$, Chebyshev's inequality applied with threshold $|\mu_{ZX}|/2$ gives

$$\mathbb{P}[A^c] \leq \frac{\sigma_{ZX}^2/n}{\mu_{ZX}^2/4} = \frac{4\sigma_{ZX}^2}{n\mu_{ZX}^2}.$$

The failure event is $B^c = \{|\bar{S}_{Z\varepsilon}| > t|\mu_{ZX}|/2\}$. Since $\mathbb{E}[\bar{S}_{Z\varepsilon}] = 0$ and $\text{Var}(\bar{S}_{Z\varepsilon}) = \sigma_{Z\varepsilon}^2/n$, Chebyshev's inequality gives

$$\mathbb{P}[B^c] \leq \frac{\sigma_{Z\varepsilon}^2/n}{t^2\mu_{ZX}^2/4} = \frac{4\sigma_{Z\varepsilon}^2}{nt^2\mu_{ZX}^2}.$$

Step 3 (Allocate the error budget and solve for t). We require the total failure probability $\mathbb{P}[A^c] + \mathbb{P}[B^c]$ to be at most δ , and we allocate half the budget to each event.

Budget for A^c . Setting $\mathbb{P}[A^c] \leq \delta/2$ requires

$$\frac{4\sigma_{ZX}^2}{n\mu_{ZX}^2} \leq \frac{\delta}{2} \iff n \geq \frac{8\sigma_{ZX}^2}{\delta\mu_{ZX}^2},$$

which is exactly condition (7).

Setting $\mathbb{P}[B^c] \leq \delta/2$ and solving for t :

$$\frac{4\sigma_{Z\varepsilon}^2}{nt^2\mu_{ZX}^2} = \frac{\delta}{2} \implies t = \frac{2\sqrt{2}\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{\delta n}}.$$

By the union bound,

$$\mathbb{P}\left[\left|\hat{\beta}_{IV} - \beta\right| > t\right] \leq \mathbb{P}[A^c] + \mathbb{P}[B^c] \leq \frac{\delta}{2} + \frac{\delta}{2} = \delta. \quad \square$$

Remark 2.4 (Polynomial dependence on δ). *The bound (8) contains the factor $1/\sqrt{\delta}$, which is polynomial in $1/\delta$. Devroye et al. (2016) show that this is unavoidable: for any estimator formed from the empirical mean of i.i.d. observations with only finite variance, the $1/\sqrt{\delta}$ rate cannot be improved without additional distributional assumptions. The MoM-based estimators*

in Sections 3–4 circumvent this by operating on block means.

Remark 2.5 (The symmetric threshold is not optimal). *The choice $D = |\mu_{ZX}|/2$ was made for simplicity. It splits the distance between 0 and $|\mu_{ZX}|$ evenly, but this is not the split that minimises the error bound for given n and δ . Section 2.1 shows that treating D as a free parameter and optimising it reduces the instrument strength constant from 8 to 1 and improves the error constant by a factor of $2\sqrt{2}$.*

2.1 Optimised Threshold

The proof of Theorem 2.3 introduced a general threshold D but then fixed it symmetrically. Here we keep D free and choose it to minimise the resulting error bound symmetrically, at the cost of a more involved Step 3.

Theorem 2.6 (Standard IV concentration, optimised threshold). *Assume (A1)–(A3). Define*

$$\rho = \frac{\sigma_{ZX}^2}{\delta n \mu_{ZX}^2}.$$

For any $\delta \in (0, 1)$, if $\rho < 1$ (i.e., $n > \sigma_{ZX}^2/(\delta \mu_{ZX}^2)$), then the choice $D^ = (1 - \rho^{1/3}) |\mu_{ZX}|$ is optimal and*

$$\mathbb{P}\left[\left|\hat{\beta}_{\text{IV}} - \beta\right| > \frac{\sigma_{Z\epsilon}}{|\mu_{ZX}| \sqrt{\delta n (1 - \rho^{1/3})^{3/2}}}\right] \leq \delta. \quad (9)$$

Proof. Steps 1 and 2 are identical to those of Theorem 2.3 but with D left as a free parameter; we arrive at the Chebyshev bounds

$$\mathbb{P}[A(D)^c] \leq \frac{\sigma_{ZX}^2/n}{(|\mu_{ZX}| - D)^2}, \quad \mathbb{P}[B(D, t)^c] \leq \frac{\sigma_{Z\epsilon}^2/n}{t^2 D^2}.$$

Requiring their sum to be at most δ gives the joint constraint

$$\frac{\sigma_{ZX}^2}{n(|\mu_{ZX}| - D)^2} + \frac{\sigma_{Z\epsilon}^2}{nt^2 D^2} \leq \delta. \quad (10)$$

Step 3 (Optimise D , then solve for t). Write $D = \eta |\mu_{ZX}|$ with $\eta \in (0, 1)$. Dividing (20) through by δ and substituting $\rho = \sigma_{ZX}^2/(\delta n \mu_{ZX}^2)$:

$$\frac{\rho}{(1 - \eta)^2} + \frac{\sigma_{Z\epsilon}^2}{\delta n \mu_{ZX}^2 \eta^2 t^2} \leq 1.$$

Solving for t^2 :

$$t^2 \geq \frac{\sigma_{Z\epsilon}^2}{\delta n \mu_{ZX}^2} \cdot \frac{1}{f(\eta)}, \quad f(\eta) = \eta^2 \left(1 - \frac{\rho}{(1 - \eta)^2}\right).$$

Minimising t is equivalent to maximising f over $\eta \in (0, 1)$. Differentiating and setting $f'(\eta) = 0$:

$$f'(\eta) = 2\eta \left(1 - \frac{\rho}{(1 - \eta)^2}\right) - \frac{2\rho\eta^2}{(1 - \eta)^3} = 0.$$

Dividing through by $2\eta > 0$:

$$1 - \frac{\rho}{(1-\eta)^2} - \frac{\rho\eta}{(1-\eta)^3} = 0 \iff (1-\eta)^3 - \rho(1-\eta) - \rho\eta = 0 \iff (1-\eta)^3 = \rho.$$

Hence $\eta^* = 1 - \rho^{1/3}$, which lies in $(0, 1)$ if and only if $\rho < 1$, the stated instrument strength condition. Substituting $(1 - \eta^*)^2 = \rho^{2/3}$:

$$f(\eta^*) = (1 - \rho^{1/3})^2 \left(1 - \frac{\rho}{\rho^{2/3}}\right) = (1 - \rho^{1/3})^2 (1 - \rho^{1/3}) = (1 - \rho^{1/3})^3.$$

The minimised threshold is therefore

$$t^* = \frac{\sigma_{Z\varepsilon}}{|\mu_{ZX}| \sqrt{\delta n} (1 - \rho^{1/3})^{3/2}},$$

and the union bound gives $\mathbb{P}[|\hat{\beta}_{IV} - \beta| > t^*] \leq \delta$. $\square$

Remark 2.7 (Comparison with the symmetric bound). *As $\rho \rightarrow 0$ (strong instruments or large n), the correction factor $(1 - \rho^{1/3})^{-3/2} \rightarrow 1$ and (19) approaches $\sigma_{Z\varepsilon}/(|\mu_{ZX}| \sqrt{\delta n})$, a factor of $2\sqrt{2} \approx 2.83$ tighter than (8). As $\rho \rightarrow 1$ (approaching the boundary of instrument relevance), the factor diverges, reflecting the genuine difficulty of ratio estimation near instrument irrelevance. The instrument strength condition is eight times weaker than Theorem 2.3: $\rho < 1$ versus $\rho < 1/8$.*

2.2 Cantelli Improvement

The Chebyshev bound applied to $A(D)^c$ in the proofs above is two-sided. However, $A(D)^c = \{|\bar{S}_{ZX}| < D\}$ is a one-sided failure event: only a *downward* deviation of $\bar{S}_{ZX}$ from μ_{ZX} causes the denominator to be too small; an upward deviation makes it larger and is harmless. Cantelli's inequality exploits this one-sidedness.

Lemma 2.8 (Cantelli's inequality). *Let W be a random variable with $\mathbb{E}[W] = \mu$ and $\text{Var}(W) = \sigma^2 < \infty$. For any $\tau > 0$,*

$$\mathbb{P}[W - \mu \leq -\tau] \leq \frac{\sigma^2}{\sigma^2 + \tau^2}.$$

This is strictly smaller than the Chebyshev bound σ^2/τ^2 for all $\tau > 0$, with the greatest relative gain when $\tau \approx \sigma$.

Theorem 2.9 (Standard IV concentration, Cantelli improvement). *Assume (A1)–(A3) and $\mu_{ZX} > 0$. Write $\gamma_{IV} = \sigma_{ZX}^2/(n\mu_{ZX}^2)$ and define*

$$h(\eta) = \eta^2 \frac{\delta(\gamma_{IV} + (1-\eta)^2) - \gamma_{IV}}{\gamma_{IV} + (1-\eta)^2}, \quad \eta \in (0, 1). \quad (11)$$

If

$$n > \frac{(1-\delta)\sigma_{ZX}^2}{\delta\mu_{ZX}^2}, \quad \text{equivalently,} \quad \gamma_{IV} < \frac{\delta}{1-\delta}, \quad (12)$$

then $h(\eta) > 0$ on a non-empty interval and the estimator satisfies

$$\mathbb{P}\left[\left|\hat{\beta}_{IV} - \beta\right| > \frac{\sigma_{Z\varepsilon}}{|\mu_{ZX}| \sqrt{nh(\eta_{IV}^*)}}\right] \leq \delta, \quad (13)$$

where η_{IV}^* is the maximiser of h over its feasible domain.

Proof. We use the same two-event framework with $D = \eta\mu_{ZX}$ kept general.

Step 1 (Two-event decomposition). As in Theorem 2.3, define $A(D) = \{\bar{S}_{ZX} \geq D\}$ and $B(D, t) = \{|\bar{S}_{Z\varepsilon}| \leq tD\}$. On $A(D) \cap B(D, t)$, $|\hat{\beta}_{IV} - \beta| \leq t$.

Step 2 (Cantelli on A^c , Chebyshev on B^c).

Since $\mu_{ZX} > 0$ and $D < \mu_{ZX}$, the failure event satisfies

$$A(D)^c = \{\bar{S}_{ZX} < D\} \subseteq \{\bar{S}_{ZX} - \mu_{ZX} < -(\mu_{ZX} - D)\}.$$

This is a one-sided downward deviation, so Lemma 4.7 applies with $W = \bar{S}_{ZX}$, $\sigma^2 = \sigma_{ZX}^2/n$, and $\tau = \mu_{ZX} - D$:

$$\mathbb{P}[A(D)^c] \leq \frac{\sigma_{ZX}^2/n}{\sigma_{ZX}^2/n + (\mu_{ZX} - D)^2}.$$

The event $B(D, t)^c = \{|\bar{S}_{Z\varepsilon}| > tD\}$ is two-sided (a large residual of either sign causes failure), so Chebyshev is appropriate:

$$\mathbb{P}[B(D, t)^c] \leq \frac{\sigma_{Z\varepsilon}^2/n}{t^2 D^2}.$$

Step 3 (Constraint and optimization). Setting the total failure probability to at most δ and substituting $D = \eta\mu_{ZX}$, $\gamma_{IV} = \sigma_{ZX}^2/(n\mu_{ZX}^2)$:

$$\frac{\gamma_{IV}}{\gamma_{IV} + (1 - \eta)^2} + \frac{\sigma_{Z\varepsilon}^2}{nt^2\eta^2\mu_{ZX}^2} \leq \delta.$$

Isolating the second term and solving for t^2 :

$$t^2 \geq \frac{\sigma_{Z\varepsilon}^2}{n\mu_{ZX}^2 h(\eta)},$$

where $h(\eta)$ is defined in (21). The function $h(\eta)$ is positive only when $\delta(\gamma_{IV} + (1 - \eta)^2) > \gamma_{IV}$, i.e., $(1 - \eta)^2 > \gamma_{IV}(1 - \delta)/\delta$. The feasible region is non-empty if and only if $\gamma_{IV} < \delta/(1 - \delta)$, which is condition (22). Minimizing t over the feasible region by maximizing h and applying the union bound gives (23). $\square$

Remark 2.10 (When Cantelli helps). *Compared with the optimised Chebyshev condition ($n > \sigma_{ZX}^2/(\delta\mu_{ZX}^2)$), the Cantelli condition (22) is weaker by a factor of $1/(1 - \delta)$. At $\delta = 0.05$ this ratio is $1/0.95 \approx 1.05$, so the gain is negligible for typical significance levels. The improvement is more meaningful at moderate confidence levels: at $\delta = 0.5$ the factor reaches 2. For the standard IV estimator the Cantelli improvement is therefore primarily of theoretical interest. By contrast, for the MoR estimator (Section ??), Cantelli extends the valid range of the instrument strength parameter from $\gamma < 1/4$ to $\gamma < 1/3$, a qualitatively more substantial gain.*

Remark 2.11 (No closed form for η_{IV}^*). The first-order condition $h'(\eta) = 0$ reduces to a quartic in η with no general closed-form solution. The optimised Chebyshev version (Theorem 2.6) yielded the clean form $\eta^* = 1 - \rho^{1/3}$ because the Chebyshev bound σ^2/τ^2 produces a simpler algebraic structure than the Cantelli bound $\sigma^2/(\sigma^2 + \tau^2)$. The bound (23) must therefore be evaluated numerically for specific parameter values.

3 The Ratio-of-Medians MoM-IV Estimator

Algorithm 2: Ratio-of-Medians Estimator

Input: Sample $(Y_i, X_i, Z_i)_{i=1}^n$, number of blocks k

Output: Estimate $\hat{\beta}_{RM}$

Partition $(Y_i, X_i, Z_i)_{i=1}^n$ into k blocks $B_1, \dots, B_k$ of size $m = \lfloor n/k \rfloor$;

for $j = 1, \dots, k$ **do**

Compute the block means $\hat{S}_{ZY}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i Y_i$ and $\hat{S}_{ZX}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i X_i$;
 Compute coordinate-wise medians $\tilde{S}_{ZY} = \text{med}(\hat{S}_{ZY}^{(1)}, \dots, \hat{S}_{ZY}^{(k)})$ and
 $\tilde{S}_{ZX} = \text{med}(\hat{S}_{ZX}^{(1)}, \dots, \hat{S}_{ZX}^{(k)})$;

return $\hat{\beta}_{RM} = \frac{\tilde{S}_{ZY}}{\tilde{S}_{ZX}}$;

Theorem 3.1 (RoM concentration). Assume (A1)–(A3). Let $\delta \in (0, 1)$, $k = \lceil 8 \ln(2/\delta) \rceil$, and $m = \lfloor n/k \rfloor$. If the instrument strength condition

$$m > \frac{4\sigma_{ZX}^2}{\mu_{ZX}^2} \quad (14)$$

holds, then

$$\mathbb{P}\left[\left|\hat{\beta}_{RM} - \beta\right| > \frac{2(\sigma_{ZY} + |\beta|\sigma_{ZX})}{|\mu_{ZX}|\sqrt{m} - 2\sigma_{ZX}}\right] \leq \delta. \quad (15)$$

Proof. The proof applies the scalar MoM result (Theorem 1.4) to each coordinate separately and combines the results with a union bound.

Step 1 (MoM on each coordinate). By Theorem 1.4 applied to the sequences $(Z_i Y_i)_{i=1}^n$ and $(Z_i X_i)_{i=1}^n$,

$$\mathbb{P}\left[\left|\tilde{S}_{ZY} - \mu_{ZY}\right| > \frac{2\sigma_{ZY}}{\sqrt{m}}\right] \leq e^{-k/8}, \quad \mathbb{P}\left[\left|\tilde{S}_{ZX} - \mu_{ZX}\right| > \frac{2\sigma_{ZX}}{\sqrt{m}}\right] \leq e^{-k/8}.$$

Step 2 (Union bound). Define the events

$$A = \left\{\left|\tilde{S}_{ZY} - \mu_{ZY}\right| \leq \frac{2\sigma_{ZY}}{\sqrt{m}}\right\}, \quad B = \left\{\left|\tilde{S}_{ZX} - \mu_{ZX}\right| \leq \frac{2\sigma_{ZX}}{\sqrt{m}}\right\}.$$

By Step 1, $\mathbb{P}[A^c] \leq e^{-k/8}$ and $\mathbb{P}[B^c] \leq e^{-k/8}$. Note that A and B are not independent (both are constructed from the same data), so we use the union bound:

$$\mathbb{P}[(A \cap B)^c] = \mathbb{P}[A^c \cup B^c] \leq \mathbb{P}[A^c] + \mathbb{P}[B^c] \leq 2e^{-k/8}.$$

Setting $k = \lceil 8 \ln(2/\delta) \rceil$ ensures $2e^{-k/8} \leq \delta$.

Fix
format-
ting

Step 3 (Algebraic bound on the ratio). On $A \cap B$, we decompose

$$\hat{\beta}_{\text{RM}} - \beta = \frac{\tilde{S}_{ZY}}{\tilde{S}_{ZX}} - \frac{\mu_{ZY}}{\mu_{ZX}} = \frac{\tilde{S}_{ZY}\mu_{ZX} - \mu_{ZY}\tilde{S}_{ZX}}{\tilde{S}_{ZX} \cdot \mu_{ZX}}.$$

Adding and subtracting $\mu_{ZY}\mu_{ZX}$ in the numerator and using $\mu_{ZY} = \beta\mu_{ZX}$,

$$\tilde{S}_{ZY}\mu_{ZX} - \mu_{ZY}\tilde{S}_{ZX} = \mu_{ZX} \left[(\tilde{S}_{ZY} - \mu_{ZY}) - \beta(\tilde{S}_{ZX} - \mu_{ZX}) \right].$$

The factor μ_{ZX} cancels with the μ_{ZX} in the denominator, giving

$$\hat{\beta}_{\text{RM}} - \beta = \frac{(\tilde{S}_{ZY} - \mu_{ZY}) - \beta(\tilde{S}_{ZX} - \mu_{ZX})}{\tilde{S}_{ZX}}.$$

Taking absolute values and applying the triangle inequality to the numerator,

$$|\hat{\beta}_{\text{RM}} - \beta| \leq \frac{|\tilde{S}_{ZY} - \mu_{ZY}| + |\beta| |\tilde{S}_{ZX} - \mu_{ZX}|}{|\tilde{S}_{ZX}|}.$$

Step 4 (Substitute bounds). On event A , $|\tilde{S}_{ZY} - \mu_{ZY}| \leq 2\sigma_{ZY}/\sqrt{m}$. On event B , $|\tilde{S}_{ZX} - \mu_{ZX}| \leq 2\sigma_{ZX}/\sqrt{m}$, and by the reverse triangle inequality,

$$|\tilde{S}_{ZX}| \geq |\mu_{ZX}| - |\tilde{S}_{ZX} - \mu_{ZX}| \geq |\mu_{ZX}| - \frac{2\sigma_{ZX}}{\sqrt{m}}.$$

The instrument strength condition (14) ensures $2\sigma_{ZX}/\sqrt{m} < |\mu_{ZX}|$, so the denominator is strictly positive. Substituting,

$$|\hat{\beta}_{\text{RM}} - \beta| \leq \frac{2\sigma_{ZY}/\sqrt{m} + |\beta| \cdot 2\sigma_{ZX}/\sqrt{m}}{|\mu_{ZX}| - 2\sigma_{ZX}/\sqrt{m}} = \frac{2(\sigma_{ZY} + |\beta|\sigma_{ZX})}{|\mu_{ZX}| \sqrt{m} - 2\sigma_{ZX}}.$$

Since this holds on $A \cap B$ and $\mathbb{P}[(A \cap B)^c] \leq \delta$, the result follows. $\square$

4 The Median-of-Ratios MoM-IV Estimator

Algorithm 3: Median-of-Ratios Estimator

Input: Sample $(Y_i, X_i, Z_i)_{i=1}^n$, number of blocks k

Output: Estimate $\hat{\beta}_{\text{RM}}$

Partition $(Y_i, X_i, Z_i)_{i=1}^n$ into k blocks $B_1, \dots, B_k$ of size $m = \lfloor n/k \rfloor$;

for $j = 1, \dots, k$ **do**

Compute the block means $\hat{S}_{ZY}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i Y_i$ and $\hat{S}_{ZX}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i X_i$;
 Compute block level IV estimates $\hat{\beta}^{(j)} = \frac{\hat{S}_{ZY}^{(j)}}{\hat{S}_{ZX}^{(j)}}$;

return $\hat{\beta}_{\text{MR}} = \text{median}(\hat{\beta}^{(1)}, \dots, \hat{\beta}^{(k)})$;

Lemma 4.1 (Block-level error decomposition). *Under (1) and (A1), for each block j ,*

$$\hat{\beta}^{(j)} - \beta = \frac{\bar{\varepsilon}_j^Z}{\hat{S}_{ZX}^{(j)}}, \quad (16)$$

where $\bar{\varepsilon}_j^Z = \frac{1}{m} \sum_{i \in B_j} Z_i \varepsilon_i$ satisfies $\mathbb{E}[\bar{\varepsilon}_j^Z] = 0$.

Proof. Multiply $Y_i = \beta X_i + \varepsilon_i$ by Z_i and average over block B_j :

$$\hat{S}_{ZY}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i (\beta X_i + \varepsilon_i) = \beta \hat{S}_{ZX}^{(j)} + \bar{\varepsilon}_j^Z.$$

Dividing by $\hat{S}_{ZX}^{(j)}$ gives $\hat{\beta}^{(j)} = \beta + \bar{\varepsilon}_j^Z / \hat{S}_{ZX}^{(j)}$. By (A1) and linearity of expectation, $\mathbb{E}[\bar{\varepsilon}_j^Z] = \frac{1}{m} \sum_{i \in B_j} \mathbb{E}[Z_i \varepsilon_i] = 0$. $\square$

Theorem 4.2 (MoR concentration). *Assume (A1)–(A3). Let $\delta \in (0, 1)$, $k = \lceil 8 \ln(1/\delta) \rceil$, and $m = \lfloor n/k \rfloor$. If the instrument strength condition*

$$m \geq \frac{32\sigma_{ZX}^2}{\mu_{ZX}^2} \quad (17)$$

holds, then

$$\mathbb{P} \left[\left| \hat{\beta}_{MR} - \beta \right| > \frac{4\sqrt{2}\sigma_{Z\varepsilon}}{|\mu_{ZX}| \sqrt{m}} \right] \leq \delta. \quad (18)$$

Proof. The proof applies the MoM counting argument directly to the block-level IV estimates $\hat{\beta}^{(j)}$.

Step 1 (Two-event decomposition). By Lemma 4.1, $\left| \hat{\beta}^{(j)} - \beta \right| = \left| \bar{\varepsilon}_j^Z \right| / \left| \hat{S}_{ZX}^{(j)} \right|$. To bound this ratio, we control the numerator above and the denominator below. For a threshold $t > 0$ to be determined, define

$$A_j(D) = \left\{ \left| \hat{S}_{ZX}^{(j)} - \mu_{ZX} \right| \leq D \right\}, \quad B_j(D, t) = \left\{ \left| \bar{\varepsilon}_j^Z \right| \leq tD \right\}.$$

On $A_j(D) \cap B_j(D, t)$, the denominator satisfies $|\bar{S}_{ZX}| \geq D$ by definition of $A_j(D)$, so

$$\left| \hat{\beta}_{IV} - \beta \right| = \frac{\left| \bar{\varepsilon}_j^Z \right|}{\left| \bar{S}_{ZX} \right|} \leq \frac{tD}{D} = t.$$

Hence $\mathbb{P} \left[\left| \hat{\beta}_{MR} - \beta \right| > t \right] \leq \mathbb{P}[A_j(D)^c] + \mathbb{P}[B_j(D, t)^c]$ by the union bound. For the remainder of this proof we set $D = |\mu_{ZX}|/2$, the symmetric choice that yields the cleanest constants.

Step 2 (Chebyshev bounds). We bound each failure probability in turn.

The failure event $A_j^c = \left\{ \left| \bar{S}_{ZX}^{(j)} \right| < |\mu_{ZX}|/2 \right\}$ implies a large deviation of $\bar{S}_{ZX}^{(j)}$ from its mean. Specifically, if $\left| \bar{S}_{ZX}^{(j)} \right| < |\mu_{ZX}|/2$ then, by the reverse triangle inequality,

$$\left| \bar{S}_{ZX}^{(j)} - \mu_{ZX} \right| \geq |\mu_{ZX}| - \left| \bar{S}_{ZX}^{(j)} \right| > |\mu_{ZX}| - \frac{|\mu_{ZX}|}{2} = \frac{|\mu_{ZX}|}{2}.$$

Hence $A_j^c \subseteq \{|\bar{S}_{ZX}^{(j)} - \mu_{ZX}| > |\mu_{ZX}|/2\}$. Since $\text{Var}(\bar{S}_{ZX}^{(j)}) = \sigma_{ZX}^2/m$, Chebyshev's inequality applied with threshold $|\mu_{ZX}|/2$ gives

$$\mathbb{P}[A_j^c] \leq \frac{\sigma_{ZX}^2/m}{\mu_{ZX}^2/4} = \frac{4\sigma_{ZX}^2}{m\mu_{ZX}^2}.$$

The failure event is $B_j^c = \{|\bar{\varepsilon}_Z^{(j)}| > t|\mu_{ZX}|/2\}$. Since $\mathbb{E}[\bar{\varepsilon}_Z^{(j)}] = 0$ and $\text{Var}(\bar{\varepsilon}_Z^{(j)}) = \sigma_{Z\varepsilon}^2/m$, Chebyshev's inequality gives

$$\mathbb{P}[B_j^c] \leq \frac{\sigma_{Z\varepsilon}^2/m}{t^2\mu_{ZX}^2/4} = \frac{4\sigma_{Z\varepsilon}^2}{mt^2\mu_{ZX}^2}.$$

The instrument strength condition (17) ensures $\mathbb{P}[A_j^c] \leq 1/8$. Setting $t = 4\sqrt{2}\sigma_{Z\varepsilon}/(|\mu_{ZX}|\sqrt{m})$ gives $\mathbb{P}[B_j^c] \leq 1/8$.

Step 3 (Allocate the error budget and solve for t). We require the total failure probability $\mathbb{P}[A^c] + \mathbb{P}[B^c]$ to be at most $\frac{1}{4}$, and we allocate half the budget to each event.

Budget for A_j^c . Setting $\mathbb{P}[A_j^c] \leq 1/8$ requires

$$\frac{4\sigma_{ZX}^2}{m\mu_{ZX}^2} \leq \frac{1}{8} \iff m \geq \frac{32\sigma_{ZX}^2}{\mu_{ZX}^2},$$

which is exactly condition (17).

Setting $\mathbb{P}[B_j^c] \leq 1/8$ and solving for t :

$$\frac{4\sigma_{Z\varepsilon}^2}{nt^2\mu_{ZX}^2} \leq \frac{1}{8} \implies t \geq \frac{\sqrt{32}\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{m}}.$$

By the union bound,

$$\mathbb{P}\left[|\hat{\beta}_{\text{IV}} - \beta| > t\right] \leq \mathbb{P}[A^c] + \mathbb{P}[B^c] \leq \frac{1}{4} + \frac{1}{4} = \frac{1}{8}.$$

Step 4 (Counting argument). Define $\zeta_j = \mathbf{1}\{|\hat{\beta}^{(j)} - \beta| > t\}$. By Step 2, $\mathbb{E}[\zeta_j] \leq 1/4$. Since the blocks are disjoint, the $(\zeta_j)_{j=1}^k$ are independent. If $|\hat{\beta}_{\text{MR}} - \beta| > t$, the median of the block estimates lies outside $(\beta - t, \beta + t)$, which requires at least $k/2$ blocks to satisfy $|\hat{\beta}^{(j)} - \beta| > t$. Therefore,

$$\mathbb{P}\left[|\hat{\beta}_{\text{MR}} - \beta| > t\right] \leq \mathbb{P}\left[\sum_{j=1}^k \zeta_j \geq \frac{k}{2}\right].$$

Step 5 (Hoeffding). By Hoeffding's inequality, with overshoot $k/2 - k/4 = k/4$,

$$\mathbb{P}\left[\sum_{j=1}^k \zeta_j \geq \frac{k}{2}\right] \leq \exp\left(\frac{-2(k/4)^2}{k}\right) = e^{-k/8}.$$

Setting $k = \lceil 8\ln(1/\delta) \rceil$ gives $e^{-k/8} \leq \delta$, completing the proof. $\square$

Remark 4.3 (Logarithmic dependence on δ). *The bound (18) contains the factor $\ln(1/\delta)$, which is polynomial in $1/\delta$.*

Remark 4.4 (The symmetric threshold is not optimal). *The choice $D = |\mu_{ZX}|/2$ was made for simplicity. It splits the distance between 0 and $|\mu_{ZX}|$ evenly, but this is not the split that minimises the error bound for given n and δ . Section 4.1 shows that treating D as a free parameter and optimising it reduces the instrument strength constant from something to something else and improves the error constant by a factor of another thing.*

add the
stuff

keep
going

4.1 Optimised Threshold

The proof of Theorem 4.2 introduced a general threshold D but then fixed it symmetrically. Here we keep D free and choose it to minimise the resulting error bound symmetrically, at the cost of a more involved Step 3.

Theorem 4.5 (Standard MoR concentration, optimised threshold). *Assume (A1)–(A3). Define*

$$\rho = \frac{4\sigma_{ZX}^2}{m\mu_{ZX}^2}.$$

For any $\delta \in (0, 1)$, if $\rho < 1$ (i.e., $n > \sigma_{ZX}^2/(\delta\mu_{ZX}^2)$), then the choice $D^ = (1 - \rho^{1/3})|\mu_{ZX}|$ is optimal and*

$$\mathbb{P}\left[\left|\hat{\beta}_{\text{IV}} - \beta\right| > \frac{\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{\delta n}(1 - \rho^{1/3})^{3/2}}\right] \leq \delta. \quad (19)$$

Proof. Steps 1 and 2 are identical to those of Theorem 2.3 but with D left as a free parameter; we arrive at the Chebyshev bounds

$$\mathbb{P}[A(D)^c] \leq \frac{\sigma_{ZX}^2/n}{(|\mu_{ZX}| - D)^2}, \quad \mathbb{P}[B(D, t)^c] \leq \frac{\sigma_{Z\varepsilon}^2/n}{t^2 D^2}.$$

Requiring their sum to be at most δ gives the joint constraint

$$\frac{\sigma_{ZX}^2}{n(|\mu_{ZX}| - D)^2} + \frac{\sigma_{Z\varepsilon}^2}{nt^2 D^2} \leq \delta. \quad (20)$$

Step 3 (Optimise D , then solve for t). Write $D = \eta|\mu_{ZX}|$ with $\eta \in (0, 1)$. Dividing (20) through by δ and substituting $\rho = \sigma_{ZX}^2/(\delta n\mu_{ZX}^2)$:

$$\frac{\rho}{(1 - \eta)^2} + \frac{\sigma_{Z\varepsilon}^2}{\delta n\mu_{ZX}^2 \eta^2 t^2} \leq 1.$$

Solving for t^2 :

$$t^2 \geq \frac{\sigma_{Z\varepsilon}^2}{\delta n\mu_{ZX}^2} \cdot \frac{1}{f(\eta)}, \quad f(\eta) = \eta^2 \left(1 - \frac{\rho}{(1 - \eta)^2}\right).$$

Minimising t is equivalent to maximising f over $\eta \in (0, 1)$. Differentiating and setting $f'(\eta) = 0$:

$$f'(\eta) = 2\eta \left(1 - \frac{\rho}{(1 - \eta)^2}\right) - \frac{2\rho\eta^2}{(1 - \eta)^3} = 0.$$

Dividing through by $2\eta > 0$:

$$1 - \frac{\rho}{(1-\eta)^2} - \frac{\rho\eta}{(1-\eta)^3} = 0 \iff (1-\eta)^3 - \rho(1-\eta) - \rho\eta = 0 \iff (1-\eta)^3 = \rho.$$

Hence $\eta^* = 1 - \rho^{1/3}$, which lies in $(0, 1)$ if and only if $\rho < 1$, the stated instrument strength condition. Substituting $(1 - \eta^*)^2 = \rho^{2/3}$:

$$f(\eta^*) = (1 - \rho^{1/3})^2 \left(1 - \frac{\rho}{\rho^{2/3}}\right) = (1 - \rho^{1/3})^2 (1 - \rho^{1/3}) = (1 - \rho^{1/3})^3.$$

The minimised threshold is therefore

$$t^* = \frac{\sigma_{Z\varepsilon}}{|\mu_{ZX}| \sqrt{\delta n} (1 - \rho^{1/3})^{3/2}},$$

and the union bound gives $\mathbb{P}[|\hat{\beta}_{IV} - \beta| > t^*] \leq \delta$. $\square$

Remark 4.6 (Comparison with the symmetric bound). *As $\rho \rightarrow 0$ (strong instruments or large n), the correction factor $(1 - \rho^{1/3})^{-3/2} \rightarrow 1$ and (19) approaches $\sigma_{Z\varepsilon}/(|\mu_{ZX}| \sqrt{\delta n})$, a factor of $2\sqrt{2} \approx 2.83$ tighter than (8). As $\rho \rightarrow 1$ (approaching the boundary of instrument relevance), the factor diverges, reflecting the genuine difficulty of ratio estimation near instrument irrelevance. The instrument strength condition is eight times weaker than Theorem 2.3: $\rho < 1$ versus $\rho < 1/8$.*

4.2 Cantelli Improvement

The Chebyshev bound applied to $A(D)^c$ in the proofs above is two-sided. However, $A(D)^c = \{|\bar{S}_{ZX}| < D\}$ is a one-sided failure event: only a *downward* deviation of $\bar{S}_{ZX}$ from μ_{ZX} causes the denominator to be too small; an upward deviation makes it larger and is harmless. Cantelli's inequality exploits this one-sidedness.

Lemma 4.7 (Cantelli's inequality). *Let W be a random variable with $\mathbb{E}[W] = \mu$ and $\text{Var}(W) = \sigma^2 < \infty$. For any $\tau > 0$,*

$$\mathbb{P}[W - \mu \leq -\tau] \leq \frac{\sigma^2}{\sigma^2 + \tau^2}.$$

This is strictly smaller than the Chebyshev bound σ^2/τ^2 for all $\tau > 0$, with the greatest relative gain when $\tau \approx \sigma$.

Theorem 4.8 (Standard IV concentration, Cantelli improvement). *Assume (A1)–(A3) and $\mu_{ZX} > 0$. Write $\gamma_{IV} = \sigma_{ZX}^2/(n\mu_{ZX}^2)$ and define*

$$h(\eta) = \eta^2 \frac{\delta(\gamma_{IV} + (1-\eta)^2) - \gamma_{IV}}{\gamma_{IV} + (1-\eta)^2}, \quad \eta \in (0, 1). \quad (21)$$

If

$$n > \frac{(1-\delta)\sigma_{ZX}^2}{\delta\mu_{ZX}^2}, \quad \text{equivalently,} \quad \gamma_{IV} < \frac{\delta}{1-\delta}, \quad (22)$$

then $h(\eta) > 0$ on a non-empty interval and the estimator satisfies

$$\mathbb{P}\left[\left|\hat{\beta}_{IV} - \beta\right| > \frac{\sigma_{Z\varepsilon}}{|\mu_{ZX}| \sqrt{nh(\eta_{IV}^*)}}\right] \leq \delta, \quad (23)$$

where η_{IV}^* is the maximiser of h over its feasible domain.

Proof. We use the same two-event framework with $D = \eta\mu_{ZX}$ kept general.

Step 1 (Two-event decomposition). As in Theorem 2.3, define $A(D) = \{\bar{S}_{ZX} \geq D\}$ and $B(D, t) = \{|\bar{S}_{Z\varepsilon}| \leq tD\}$. On $A(D) \cap B(D, t)$, $|\hat{\beta}_{IV} - \beta| \leq t$.

Step 2 (Cantelli on A^c , Chebyshev on B^c).

Since $\mu_{ZX} > 0$ and $D < \mu_{ZX}$, the failure event satisfies

$$A(D)^c = \{\bar{S}_{ZX} < D\} \subseteq \{\bar{S}_{ZX} - \mu_{ZX} < -(\mu_{ZX} - D)\}.$$

This is a one-sided downward deviation, so Lemma 4.7 applies with $W = \bar{S}_{ZX}$, $\sigma^2 = \sigma_{ZX}^2/n$, and $\tau = \mu_{ZX} - D$:

$$\mathbb{P}[A(D)^c] \leq \frac{\sigma_{ZX}^2/n}{\sigma_{ZX}^2/n + (\mu_{ZX} - D)^2}.$$

The event $B(D, t)^c = \{|\bar{S}_{Z\varepsilon}| > tD\}$ is two-sided (a large residual of either sign causes failure), so Chebyshev is appropriate:

$$\mathbb{P}[B(D, t)^c] \leq \frac{\sigma_{Z\varepsilon}^2/n}{t^2 D^2}.$$

Step 3 (Constraint and optimization). Setting the total failure probability to at most δ and substituting $D = \eta\mu_{ZX}$, $\gamma_{IV} = \sigma_{ZX}^2/(n\mu_{ZX}^2)$:

$$\frac{\gamma_{IV}}{\gamma_{IV} + (1 - \eta)^2} + \frac{\sigma_{Z\varepsilon}^2}{nt^2\eta^2\mu_{ZX}^2} \leq \delta.$$

Isolating the second term and solving for t^2 :

$$t^2 \geq \frac{\sigma_{Z\varepsilon}^2}{n\mu_{ZX}^2 h(\eta)},$$

where $h(\eta)$ is defined in (21). The function $h(\eta)$ is positive only when $\delta(\gamma_{IV} + (1 - \eta)^2) > \gamma_{IV}$, i.e., $(1 - \eta)^2 > \gamma_{IV}(1 - \delta)/\delta$. The feasible region is non-empty if and only if $\gamma_{IV} < \delta/(1 - \delta)$, which is condition (22). Minimizing t over the feasible region by maximizing h and applying the union bound gives (23). $\square$

Remark 4.9 (When Cantelli helps). Compared with the optimised Chebyshev condition ($n > \sigma_{ZX}^2/(\delta\mu_{ZX}^2)$), the Cantelli condition (22) is weaker by a factor of $1/(1 - \delta)$. At $\delta = 0.05$ this ratio is $1/0.95 \approx 1.05$, so the gain is negligible for typical significance levels. The improvement is more meaningful at moderate confidence levels: at $\delta = 0.5$ the factor reaches 2. For the standard IV estimator the Cantelli improvement is therefore primarily of theoretical interest. By contrast, for the MoR estimator (Section ??), Cantelli extends the valid range of the instrument strength parameter from $\gamma < 1/4$ to $\gamma < 1/3$, a qualitatively more substantial gain.

Remark 4.10 (No closed form for η_{IV}^*). *The first-order condition $h'(\eta) = 0$ reduces to a quartic in η with no general closed-form solution. The optimised Chebyshev version (Theorem 2.6) yielded the clean form $\eta^* = 1 - \rho^{1/3}$ because the Chebyshev bound σ^2/τ^2 produces a simpler algebraic structure than the Cantelli bound $\sigma^2/(\sigma^2 + \tau^2)$. The bound (23) must therefore be evaluated numerically for specific parameter values.*

References

- Devroye, L., Lerasle, M., Lugosi, G. & Oliveira, R. I. (2016). Sub-gaussian mean estimators.
- Hoeffding, W. (1963). Probability inequalities for sums of bounded random variables. *Journal of the American statistical association*, 58(301), 13–30.
- Lugosi, G. & Mendelson, S. (2019). Mean estimation and regression under heavy-tailed distributions: A survey. *Foundations of Computational Mathematics*, 19(5), 1145–1190.